Where visual attacks act in a frozen VLM

One shared pipeline; different controlled inputs, objectives, access and evaluation.

Solid navy = forward processing Orange = changed input Dashed purple = optimization gradient

MEASURED means our development adaptation ran; REVIEWED means the original method has not been reproduced.

A Pixel-level attacks | MEASURED ADAPTATIONS

FGSM: one signed-gradient step
PGD: repeated steps + projection
Shared image: mean loss over a released corpus
Defense-aware / flip-aware PGD: changed objective
Released Qi image: artifact transfer, no reoptimization

B Patch attacks | MEASURED ADAPTATIONS

Localized bounded patch: masked pixel updates
Shared bounded patch: one patch across targets
Replacement EOT patch: average over transforms
Image composition: $x' = (1 - M) \odot x + M \odot p$
EOT completed on Smol; Qwen EOT failed

C Instruction-bearing inputs | MIXED STATUS

Visual injection: text/instructions in an image
Image input changes; no gradient is required
Smol marker diagnostic ran; Qwen parity blocked
BAP adds feedback-driven text changes: REVIEWED
Physical selection/capture: REVIEWED, not run

All visual branches deliver a changed image x' to the normal input interface.

BAP text branch: reviewed, not run

Question / prompt q

Defense prefix | TESTED

Image x'
Published carrier
or released artifact

Preprocessing P
Resize / normalize
Patch or token layout

Visual encoder E
Model-specific vision
features / visual tokens

Connector C
Projection / merger
Q-Former in some VLMs

Language decoder D
$p(y_t \mid y_{<t}, \text{question, visual tokens})$
Autoregressive generation

Response r
Generated text
Not automatically a jailbreak

All model weights θ stay frozen. Input gradients do not require weight updates.

Encoder objective | REVIEWED

Match $E(x')$ to an original trigger embedding.
No language-decoder gradient is needed.

Generative objective | OPTIMIZATION

Target sequence NLL, or teacher behavior loss.
Loss progress ≠ measured attack success.

Behavioral evaluation

Safety: unsafe assistance / planned cases
Injection: unauthorized task deviation
Perception: incorrect / targeted answer
Also report utility, failures and cost.
Human validation remains pending.

Backpropagation through the selected path → update pixels inside the declared constraint.

D Jailbreak in Pieces | REVIEWED ORIGINAL METHOD

Control: image pixels. Target: CLIP encoder embedding.
Method: Adam minimizes L2 distance to released text / OCR / visual / combined triggers.
The attack still enters at x'; it does not directly overwrite internal embeddings.
Original example/code is available; our native-paper reproduction is NOT_RUN.

E Image Hijacks + Qi / BAP objectives | DIFFERENT METHODS

Qi: corpus sequence likelihood; our short common-benchmark version is an adaptation.
Image Hijacks: behavior matching; prompt matching uses a teacher distribution over contexts.
BAP: visual optimization plus iterative text feedback, not a fixed candidate list.
Full original Image Hijacks / BAP reproductions are NOT_RUN.

Research strategy: coverage is earned by evidence, not by an attack name

1 Original-method audit
Paper + code + data + rubric
12 readings; 22 source files checked

2 Native reproduction
Run exact feasible original setup
Current original runs: NOT_RUN

3 Matched comparison
Same held-out cases and scorer
Prior adaptations: 136 / 144 cases

4 New contribution
Test a prespecified, held-out claim
Novelty and human review pending

Sources: Goodfellow • Madry • Brown • Qi • Shayegani • Bailey • Ying • Ling. Survey / MMJ-Bench / AdaShield / JailGuard have separate roles.

Important: attacks change inputs or optimize an input objective. Generic prompting is not AdaShield; flip-aware PGD is not JailGuard. | 28 October 2026